Table 1: Cell-type classification performance (Macro-F1, %). The best value in each column is shown in bold. Δ is measured in percentage points (pp).

Representation	Kang ctrl	Kang stim	PBMC
Original embedding	82.75	84.52	82.68
SAE: MSE	83.01	84.19	81.79
SAE: MSE + KL ($\lambda = 0.3$)	**84.18**	83.62	81.61
SAE: MSE + KL (best λ)	**84.18**	**85.26**	**83.34**
Δ vs. original (best λ)	$+1.42$	$+0.73$	$+0.67$